

PHYS 143 – Mechanics and Thermal Physics

Newton's Laws and Forces

Reference and practice deck

Supporting material cut from the live lecture. Nothing here is new physics.

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How to use this deck

Labels

REFERENCE keep for later use: equation anatomy, toolkits, the master procedure, connect boxes.

PRACTICE use to develop the skill: the extra example.

Where things are

Part A Contact 1: anatomy of net force and weight, Toolkit 1, chain summary.

Part B Contact 2: anatomy of the second and third laws, Toolkits 2 to 5, the pushed-box example, chain summary.

Part C Contact 3: friction and circular anatomy, Toolkits 6 and 7, the full friction-misconception slide, the connections.

Part D the full-deck master procedure, seven toolkits and key ideas.

The live lecture is `lecture03_forces.pdf`; its master toolkit is the one procedure to use first. Slide titles that mention a Toolkit refer to the seven toolkits listed in Part D. Example numbers in this deck are those of the full deck (`lecture03_forces_FULL_backup.pdf`). In the live deck the pushed box is the answer to the opening poll of Contact 2, so the live Examples run from 1 to 8 without it.

EQUATION – net force

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \dots$$

Meaning. several forces on one object act as their vector sum.

Variables. $\vec{F}_i$ are all the forces *on* the object.

Units. newtons, N.

Interpretation. forces at an angle partly cancel; add by components, not by magnitudes.

Example. the three forces earlier: 128 N at 141° .

EQUATION – weight

$$w = mg$$

Meaning. the gravitational force of the Earth on an object, toward the Earth's centre.

Variables. m mass; $g = 9.8 \text{ m/s}^2$ near the Earth's surface.

Units. $\text{kg} \times \text{m/s}^2 = \text{N}$.

Interpretation. weight depends on the place (g); mass does not.

Example. 2.0 kg: 19.6 N on Earth, 3.2 N on the Moon.

 TOOLKIT 1 – Free-body diagrams

1. **ISOLATE** one body and draw it as a dot.
2. **ASK** “Which other objects interact with this body?” Everything touching it, and the Earth.
3. **DRAW** one arrow per interaction, starting at the dot, in the direction the force acts *on the body*. Label each with its agent: “Earth on box (w)”, “table on box (n)”, and so on.
4. **AXES** are added afterwards (Toolkit 2).

RULE: only forces acting *on* the body, each with a named agent.

WATCH OUT: do not draw the net force, ma , the velocity or forces the body exerts on other objects.

CHECK: can I name an agent for every arrow, and have I missed any object touching the body?

Interaction**Force** (vector, named by agent)**Net force****First law:** $\sum \vec{F} = 0 \Rightarrow \vec{a} = 0$ **You should be able to**

- name a force by who exerts it on whom;
- find a net force by components;
- state and apply the first law;
- distinguish mass from weight;
- draw a correct free-body diagram.

Connect: Lectures 4, 7 and 8

The forces on your diagram decide which ones do work (Lecture 4), how momentum changes (Lecture 7), and which produce torque (Lecture 8).

EQUATION – second law

$$\sum \vec{F} = m\vec{a} \quad \Longleftrightarrow \quad \sum F_x = ma_x, \quad \sum F_y = ma_y$$

Meaning. the net force on an object produces an acceleration in the same direction, inversely proportional to the mass.

Variables. $\sum \vec{F}$ net force on the object; m its mass; $\vec{a}$ its acceleration.

Units. $\text{N} = \text{kg m/s}^2$.

Interpretation. forces cause *changes* of velocity, not velocity itself; $a = 0$ is equilibrium.

Example. the pushed box: $20 \text{ N}/40 \text{ kg} = 0.50 \text{ m/s}^2$.

Predict first

A 20 N force acts on a 5.0 kg cart on a smooth floor. Predict the size of a , and what happens if m is doubled.

 TOOLKIT 2 – Choosing axes and resolving forces

CHOOSE axes to simplify the physics, not automatically horizontal and vertical. Put one axis along the acceleration (or the motion) and the other perpendicular to it. On an incline: x parallel to the surface, y perpendicular.

RESOLVE each force that is not along an axis into components, using the Week 1 toolkit: the angle from an axis, sketch the signs first.

SIGNS: a component is positive along the positive axis direction, negative against it.

WATCH OUT: a force with no component along an axis does not appear in that axis equation.

CHECK: do the components, recombined, give back the original force?

✂ TOOLKIT 3 – Applying $\sum \vec{F} = m\vec{a}$ by components

1. Draw the free-body diagram (Toolkit 1).
2. Choose the axes and resolve forces (Toolkit 2).
3. Write one equation per axis: $\sum F_x = ma_x$, $\sum F_y = ma_y$, *before* inserting numbers.
4. Count unknowns and equations; they must match.
5. Solve algebraically, then substitute with units. For equilibrium put $a = 0$.
6. Evaluate: sign, units, size, a limiting case.

WATCH OUT: n is found from the equation perpendicular to the surface; do not assume $n = mg$.

CHECK: does the direction of $\vec{a}$ agree with the direction of the net force in the diagram?

Question

A worker pushes a 40 kg box on a smooth floor with a constant horizontal force of 20 N. Find its acceleration.

Predict first

Sketch the FBD. Which forces appear in the x equation? Roughly: less than or more than 1 m/s^2 ?

Question

A worker pushes a 40 kg box on a smooth floor with a constant horizontal force of 20 N. Find its acceleration.

Predict first

Sketch the FBD. Which forces appear in the x equation? Roughly: less than or more than 1 m/s^2 ?

Set up: $+x$ along the push. The forces $\vec{w}$ and $\vec{n}$ are vertical and cancel. **Execute:**

$$\sum F_x = 20 = 40 a_x \Rightarrow a_x = 0.50 \text{ m/s}^2.$$

Evaluate: positive (along the push), units m/s^2 , less than 1 m/s^2 . ✓

EQUATION – third law

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$$

Meaning. forces always come in pairs: if A pushes on B, B pushes on A.

Variables. $\vec{F}_{A \text{ on } B}$ is the force exerted *by* A *on* B.

Units. N for each member.

Interpretation. equal in size, opposite in direction, same type, acting on *different* objects, so never cancelling in one diagram.

Example. the book on the table, next slides.

Predict first

A small car and a heavy truck collide head-on. Which experiences the larger force? Which has the larger acceleration?

 TOOLKIT 4 – Identifying third-law pairs

1. **NAME** the force: “A on B”. 2. **FLIP** it: the partner is “B on A”, the same type of force.
 3. **LOCATE** the partner: it acts on the *other* object, so it appears in that object’s free-body diagram, not in yours.
 4. **COMPARE** sizes: equal magnitude, opposite direction. The accelerations differ if the masses differ.
- RULE:** a pair is never two forces on the same object.
- WATCH OUT:** two forces that merely have equal size in a diagram are not a pair unless they are “A on B” and “B on A”.
- CHECK:** does each member of the pair act on a different object?

 TOOLKIT 5 – Connected objects

1. **ISOLATE** each body and draw its own free-body diagram.
2. **CONSTRAINT**: state the link: same acceleration magnitude, same rope tension (ideal rope and pulley).
3. **AXES**: choose each body's axis along its own motion, with signs consistent with the constraint.
4. **WRITE** one $\sum F = ma$ equation per body along its motion.
5. **SOLVE** the simultaneous equations for a and T .

WATCH OUT: drawing one diagram for the whole system and forgetting that internal tensions appear only when you separate the bodies.

CHECK: treat the whole system once: the external force equals the total mass times a . Check limiting cases.

FBD**Axes and components**

$$\sum F_x = ma_x, \quad \sum F_y = ma_y$$

**Solve and evaluate****You should be able to**

- apply the second law by components;
- solve equilibrium problems;
- identify third-law pairs and explain why they do not cancel;
- solve two-body connected problems.

Connect: Lectures 4, 5 and 7

Energy methods (Lectures 4 and 5) give speeds without solving for a ; the momentum form $\sum \vec{F} = d\vec{p}/dt$ is the general second law (Lecture 7).

EQUATION – static friction

$$0 \leq f_s \leq \mu_s n$$

Meaning. friction that prevents sliding; it adjusts to the other forces.

Variables. n normal force; μ_s coefficient of static friction (no units).

Units. N; μ_s dimensionless.

Interpretation. $\mu_s n$ is the *maximum* value, reached just before sliding.

Example. crate: $f_s = 150$ N although $\mu_s n = 196$ N.

EQUATION – kinetic friction

$$f_k = \mu_k n$$

Meaning. friction on surfaces that slide over each other.

Variables. n normal force; μ_k coefficient of kinetic friction (no units).

Units. N; μ_k dimensionless.

Interpretation. a fixed value, directed against the sliding; usually $\mu_k < \mu_s$.

Example. crate sliding: $0.40 \times 392 = 157$ N.

$f_s = \mu_s n$ is the maximum static friction, not its value

Static friction adjusts itself to match the other forces, up to $\mu_s n$.

Example: a 40 kg crate on a floor with $\mu_s = 0.50$. The maximum static friction is $0.50 \times 392 = 196 \text{ N}$.

If you push with 150 N and the crate does not move, then $f_s = 150 \text{ N}$, *not* 196 N.

Two ways to get it wrong

- (1) Using $f_s = \mu_s n$ when the object is not about to slide.
- (2) Using $\mu_s n$ when the object is already sliding: then use $f_k = \mu_k n$.

 TOOLKIT 6 – Friction and the normal force

1. **NORMAL FORCE:** use the equation perpendicular to the surface to find n when n is not already known; do not assume $n = mg$.
2. **IS IT SLIDING?** If not, friction is whatever is needed (static), and must satisfy $f_s \leq \mu_s n$. Test with $\sum F = 0$.
3. **IF SLIDING,** $f_k = \mu_k n$, directed against the relative sliding of the surfaces.
4. Write $\sum F = ma$ along the motion (Toolkit 3).

WATCH OUT: friction does not always oppose motion relative to the ground; it opposes sliding relative to the *surface*.

CHECK: is $f_s \leq \mu_s n$ satisfied if the object is at rest?

EQUATION – net radial force

$$\sum F_{\text{radial}} = \frac{mv^2}{R} = m\omega^2 R$$

Meaning. the net inward force that bends the path of the object.

Variables. m mass; v speed; R radius; ω angular speed.

Units. $\text{kg m}^2/\text{s}^2/\text{m} = \text{N}$.

Interpretation. points to the centre; supplied by real forces (friction, tension, gravity, normal force).

Example. the flat curve: 5400 N of friction.

EQUATION – maximum speed on a flat curve

$$v_{\text{max}} = \sqrt{\mu_s g R}$$

Meaning. the fastest safe speed when friction supplies the inward force.

Variables. μ_s static friction coefficient; g ; R radius.

Units. m/s.

Interpretation. independent of the mass; a wet road (small μ_s) lowers it.

Example. $\mu_s = 0.80$, $R = 50.0 \text{ m}$: 19.8 m/s.

 TOOLKIT 7 – Circular dynamics

1. **FBD** of the object with real forces only.
2. **RADIAL AXIS:** positive toward the centre.
3. **WRITE** $\sum F_{\text{radial}} = mv^2/R$ (or $m\omega^2 R$), and the equation for the perpendicular direction (for a flat curve, $n = mg$).
4. **IDENTIFY** which real force supplies the inward net force.
5. **LIMIT:** the maximum speed occurs when the supplying force reaches its maximum value.

WATCH OUT: do not draw a “centrifugal” or “centripetal” force.

CHECK: does the net force point toward the centre, and is the size reasonable compared with mg ?

Friction

Friction removes mechanical energy and warms the surfaces: work and energy in Lectures 4 and 5, heat and thermal energy in Lectures 9 to 11.

Inclines and ramps

The same slope problem is solved much more simply by energy conservation in Lecture 5, with no free-body axes.

Circular dynamics

$\sum F_{\text{radial}} = mv^2/R$ becomes the dynamics of rotating bodies in Lecture 8: torque and angular acceleration.

Physical meaning

The free-body diagram and component equations are the method you will keep: later topics change *which* forces appear, not how you set up the problem.

MASTER PROCEDURE – FORCE PROBLEMS

1. **IDENTIFY** Which body or bodies, and what is asked?
 2. **ISOLATE** One body at a time.
 3. **DRAW FBD(S)** Forces on the body, each named by agent (Toolkit 1).
 4. **CHOOSE AXES** Along the acceleration or the surface (Toolkit 2).
 5. **PREDICT** Signs, direction, trend, rough size.
 6. **WRITE $\sum F = ma$ BY COMPONENTS** One equation per axis per body (Toolkits 3 to 7).
 7. **SOLVE** Algebra first, then numbers with units.
 8. **EVALUATE** Units, sign, direction, limiting cases, back to the FBD.
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2

SET UP $\leftrightarrow$ steps 3–5

EXECUTE $\leftrightarrow$ steps 6–7

EVALUATE $\leftrightarrow$ step 8

Physical meaning

The FBD is the bridge:

situation $\rightarrow$ FBD $\rightarrow$

$$\sum F_x = ma_x, \sum F_y = ma_y$$

$\rightarrow$ motion.

 THE SEVEN TOOLKITS

- | | | |
|---|---|--|
| 1 | Free-body diagrams | isolate one body; interactions; forces on the body, named by agent |
| 2 | Choosing axes and resolving forces | axes to simplify; Week 1 components; incline axes |
| 3 | Applying $\sum F = ma$ | one equation per axis; equilibrium as $a = 0$ |
| 4 | Third-law pairs | "A on B" and "B on A"; different bodies |
| 5 | Connected objects | one FBD per body; same a ; ideal rope and pulley |
| 6 | Friction and the normal force | find n ; static or kinetic; $f_s \leq \mu_s n$, $f_k = \mu_k n$ |
| 7 | Circular dynamics | radial axis; real forces only; $\sum F_{\text{radial}} = mv^2/R$ |

The Master Procedure sits above these. Week 1 and Week 2 toolkits (components, signs, equation selection) are reused, not renamed.

Key ideas

- Every force has an agent; name it.
- The free-body diagram contains only forces on the body.
- $\sum \vec{F} = m\vec{a}$ holds component by component.
- Third-law pairs act on different bodies.
- $f_s \leq \mu_s n$ is a limit; $f_k = \mu_k n$ is fixed.
- Circular motion needs an inward net force from real forces.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slides.